Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For: Collecting and Reading Data

Every day we are surrounded by facts and figures: how many goals a team scored, the marks in a test, the number of people in your street, the price of food in the market. These facts and figures, collected together, are called data. Statistics is the branch of mathematics that deals with collecting, organising, and reading data so that it can be understood and used.

What is statistics for? Statistics has several clear purposes. First, it helps us collect information in a careful and fair way, instead of guessing. Second, it helps us organise a large amount of information so that it is not a confusing jumble. Third, it helps us compare things — for example, which fruit is most liked by students. Fourth, it helps us find patterns and trends, such as whether rainfall increases in a certain month. Finally, it helps us make decisions and predictions — a shopkeeper uses data on what people buy to decide what to stock, and a government uses data on population to plan schools and hospitals.

How is data collected? Data can be gathered by counting (how many books on a shelf), by measuring (the heights of your classmates), by asking questions (a survey or questionnaire), by observing (watching traffic pass a road), or by reading existing records (a school register).

How is data tabulated? Once data is collected, it is put into a table. A table arranges the information neatly in rows and columns so that you can see at a glance how often each item occurs. The number of times an item appears is its frequency. A table that shows each item together with its frequency is called a frequency table.

Worked example 1 — Collecting and tabulating data

A class wants to know the favourite sport among 15 students. Each student names one sport. The answers collected are:

Football, Tennis, Football, Volleyball, Football, Tennis, Basketball, Football, Volleyball, Football, Tennis, Football, Basketball, Tennis, Football

Step 1 — Collect. The data is gathered by asking each student one question. The answers above are the raw data.

Step 2 — Tally. To organise it, we make a tally table. For each answer we draw one stroke. Every fifth stroke is drawn across the previous four to make counting easier (a group of five, like IIII).

Sport	Tally	Frequency
Football	IIII II	7
Tennis	IIII	4
Volleyball	II	2
Basketball	II	2

Step 3 — Interpret. Reading the table, Football has the highest frequency (7), so Football is the most popular sport in the class. Basketball and Volleyball are equally popular, each with 2.

Worked example 2 — Reading and comparing data in a table

The table below shows the marks scored by students in a Mathematics test.

Marks	10	20	30	40	50
Frequency	3	5	8	6	2

Read the table to answer: (a) How many students scored 40? (b) Which mark was scored by most students? (c) How many students took the test?

(a) The frequency below 40 is 6, so 6 students scored 40.
(b) The largest frequency is 8, under the mark 30, so most students scored 30.
(c) Add all the frequencies: $3 + 5 + 8 + 6 + 2 = 24$ students.

Practice

1. State three purposes of statistics, and name one way in which data can be collected.
2. A teacher asked 12 students their favourite colour. The answers were: Red, Blue, Red, Green, Blue, Red, Red, Blue, Green, Red, Blue, Red.
   (a) Make a frequency table of these colours.
   (b) Which colour is the most popular?
3. The frequency table below shows the number of siblings of some students.

Number of siblings	**0**	**1**	**2**	**3**	**4**
Frequency	4	7	9	5	2

   (a) How many students have exactly 2 siblings?
   (b) How many students in total were asked?
   (c) Which number of siblings is the most common?

Section 2: Making Sense of Numbers with Tables

When we collect a lot of numbers, they can be hard to read. Statistics is the way we organise those numbers so they start to tell us something useful. One of the most helpful tools for doing this is a table.

A table takes rows of information and lines them up in columns, so you can see at a glance what is there. For example, suppose a class counted how many books each student read in a month. Instead of listing twenty separate numbers, we can put them in a table like this:

Number of books read	**2**	**3**	**4**	**5**
Number of students	4	7	6	3

To read a table, you ask yourself simple questions and let the numbers answer them.

Question: How many students read exactly 3 books? Look down the "Number of books read" row until you find 3. The number underneath it, 7, tells you that 7 students read 3 books.

Question: How many students were counted altogether? Add every number in the bottom row: $4 + 7 + 6 + 3 = 20$ students.

Question: How many students read 4 or 5 books? Add the students in those two columns: $6 + 3 = 9$ students.

So a table lets you compare groups, find totals, and answer questions quickly. The columns and rows are your signposts — always read the headings first so you know what each number is counting.

Example — Favourite fruits

A shop asked its customers which fruit they liked most and recorded the answers in this table:

Fruit	**Apple**	**Banana**	**Mango**	**Orange**
Customers	12	8	15	5

Let's read a few facts from it.

Step 1: Which fruit was liked most?
Look for the biggest number in the "Customers" row. The biggest is 15, which sits under Mango. So mango was the favourite fruit.

Step 2: How many customers chose banana or orange?
Add the customers for those two fruits: $8 + 5 = 13$ customers.

Step 3: How many customers answered altogether?
Add the whole "Customers" row: $12 + 8 + 15 + 5 = 40$ customers.

Step 4: How many more customers chose apple than orange?
The apple column has 12 and the orange column has 5. The difference is $12 - 5 = 7$ more customers chose apple.

Notice how each answer came straight from reading the numbers in the table. The headings told us what the numbers meant, and the numbers answered our questions.

To interpret a table confidently, use these three steps every time:

1. Read the headings. The row and column headings tell you what the numbers are counting. Without them, numbers have no meaning.
2. Find the right column or row. Put your finger on the heading you are interested in and follow it to the number you need.
3. Do the arithmetic the question asks for. Sometimes you just read one number. Sometimes you add columns together, or subtract one from another to find a difference.

A table is a summary: it takes a messy list of data and turns it into answers you can find. Once you can read a table, you can interpret statistical data — which simply means turning the numbers back into a clear description of what they show.

Practice

1. A sports club asked its members which sport they play most often. The results are shown in this table:

Sport	**Football**	**Basketball**	**Tennis**	**Swimming**
Members	18	10	6	14

   (a) How many members play tennis?
   (b) How many members were asked altogether?
   (c) Which sport is played by the most members?
   (d) How many more members play football than basketball?

2. A farmer recorded how many baskets of oranges each tree gave. The table shows the results:

Tree	**Tree A**	**Tree B**	**Tree C**	**Tree D**
Baskets	9	12	7	15

   (a) Which tree gave the fewest baskets?
   (b) How many baskets did Tree B and Tree C give together?
   (c) How many more baskets did Tree D give than Tree A?

Check your work

Attempt every question before you read this page.

What Statistics Is For: Collecting and Reading Data

1. Answers may vary. Three purposes of statistics include: collecting information carefully, organising data into a neat form, comparing things, finding patterns or trends, and making decisions or predictions. One way data can be collected: counting, measuring, asking questions (survey), observing, or reading existing records.

   Statistics helps us collect, organise, compare, find patterns, and make decisions. Data can be gathered by counting, measuring, asking questions, observing, or reading records — any one of these is correct.
2. (a) Frequency table: Red — 6, Blue — 4, Green — 2.
   (b) Red is the most popular, with frequency 6.

   Count each colour. Red appears at positions 1, 3, 6, 7, 10 and 12, so 6 times. Blue appears at 2, 5, 8, 11, so 4 times. Green appears at 4 and 9, so 2 times. The table shows Red (6), Blue (4), Green (2). The highest frequency is 6, so Red is most popular.
3. (a) 9 students.
   (b) $4 + 7 + 9 + 5 + 2 = 27$ students.
   (c) 2 siblings, because it has the highest frequency of 9.

   (a) Read the frequency below the "2" column: 9.
   (b) Add all frequencies: $4 + 7 + 9 + 5 + 2 = 27$.
   (c) The largest frequency is 9, which sits under "2 siblings", so the most common number of siblings is 2.

Making Sense of Numbers with Tables

1. (a) 6 members play tennis. (b) 18 + 10 + 6 + 14 = 48 members. (c) Football, with 18 members. (d) 18 − 10 = 8 more members play football than basketball.

   (a) Read the number under Tennis: it is 6.
   (b) Add the whole Members row: 18 + 10 + 6 + 14 = 48.
   (c) Find the largest number in the Members row: 18 is the biggest, and it is under Football.
   (d) Subtract the smaller from the larger: 18 − 10 = 8.
2. (a) Tree C, with 7 baskets. (b) 12 + 7 = 19 baskets. (c) 15 − 9 = 6 more baskets.

   (a) The smallest number in the Baskets row is 7, under Tree C.
   (b) Add the baskets for Tree B and Tree C: 12 + 7 = 19.
   (c) Subtract Tree A's baskets from Tree D's: 15 − 9 = 6.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-04e5e44e	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____